

CARLOS MICHAEL MARPAUNG

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Engineer with full-stack experience across backend, frontend, and database development using PHP (Laravel), Django, React, and React Native, plus enterprise SQL Server work building regulatory reporting systems (OJK Antasena, SLIK, LPS, Basel III) for the banking sector. Comfortable owning a feature end-to-end from database design and API development to UI implementation and manual QA/testing. Currently building the university's academic LMS Portal (frontend, backend, and database) at Podomoro University.

TECHNICAL SKILLS

Backend & Full-Stack: PHP (Laravel), Python (Django), C# (.NET), JavaScript (React, React Native)

Mobile: Dart (Flutter)

Database & Data: SQL Server, MySQL, PostgreSQL, Stored Procedures, CTEs, Window Functions, Query Optimization, ETL, Data Validation, Relational Schema Design

Testing & Practices: Manual Test Case Documentation, Git, Postman, RESTful API Development, Agile Collaboration

WORK EXPERIENCE

Podomoro University

Aug 2026 – Present

IT Development Staff

- Develop the university's academic LMS Portal end-to-end, covering frontend, backend, and database using PHP (Laravel), supporting core operations such as course scheduling, materials, tasks, and scoring.
- Design and maintain relational database schemas and queries for the LMS Portal in coordination with academic and administrative stakeholders.
- Author manual test case documentation for LMS modules (e.g., Schedule Exchange, Manage Material Course, Manage Task, Manage Score) as part of the team's QA process before release.

PT Bank Oke Indonesia Tbk (OK Bank)

Aug 2025 – Jul 2026

SQL Developer

- Developed and maintained 50+ SQL Server stored procedures using CTEs, window functions, and complex multi-table joins to support regulatory reporting for OJK Antasena, SLIK, LPS Single Customer View (SCV), and Basel III liquidity reports (LCR & NSFR).
- Built and enhanced SQL Server-based ETL pipelines supporting daily and monthly reporting processes, reducing manual intervention and improving data processing efficiency for enterprise banking reporting.
- Validated millions of banking records across complex reporting datasets, ensuring data accuracy, consistency, and compliance with OJK, Bank Indonesia (BI), and LPS reporting requirements.
- Optimized complex SQL queries and resolved production data discrepancies through query refactoring and execution plan analysis, improving reporting performance by approximately 30 - 40%.
- Provided production support for regulatory reporting systems by troubleshooting data issues, documenting stored procedure logic, and collaborating with cross-functional teams to ensure timely and reliable report submissions.

PT Dimensi Kreasi Nusantara

Aug 2024 – Dec 2024

Software Engineer (Internship)

- Developed core features for the Dots Mobile Application, a digital banking-style financial platform, using React Native, including authentication, savings, deposits, withdrawals, and loan management.
- Built and maintained 20+ RESTful API endpoints using Django REST Framework and PostgreSQL to support secure financial transactions and user account management.
- Designed backend modules for transaction processing, loan due-date tracking, and account management with validation, authentication, and error handling.
- Debugged frontend-backend integration issues between React Native and Django REST Framework, improving reliability of end-to-end transaction flows.
- Collaborated with cross-functional teams using Git-based workflows to develop, test, and deliver features in an Agile development environment.

FEATURED PROJECTS

Book Recommendation System (RNN/LSTM)

Dec 2024 – Jun 2025

Python · LSTM · React · Django REST Framework · Supabase

- Built an ML-powered book recommendation engine using RNN (LSTM), achieving 87% recommendation accuracy on 10,000+ book records sourced from the Google Books API.
- Trained and evaluated the model in Python and exposed predictions via a Django REST Framework API consumed by a React frontend backed by Supabase.

Workshop Management System

Dec 2024 – Mar 2025

PHP · Laravel · Livewire · Tailwind CSS · MySQL

- Built a Laravel/Livewire-based POS and inventory management system for Batik Tegalana with real-time stock tracking, customer reservations, and educational content management.
- Centralized 5+ operational workflows into a single admin dashboard, reducing manual data entry and improving daily transaction tracking.

E-Commerce Management System

May 2024 – Jul 2024

C# · .NET · SQL Server

- Engineered a full-stack e-commerce platform supporting dual roles (Admin & Customer) with 10+ stored procedures for transaction processing and reporting.
- Built an admin dashboard tracking revenue, stock levels, and customer data across 300+ SKUs.

More projects (BAAK Management System, Library Management System, Course SkillUp) available on carlos-marpaung-portfolio.vercel.app

EDUCATION

Institut Teknologi Del

Sep 2022 – Jul 2025

Diploma 3 in Information Technology, GPA 3.50/4.00

- Awarded the Pelangi Scholarship – PT Toba Pulp Lestari (2022–2025), a full scholarship covering tuition, dormitory, and living expenses, granted based on academic excellence and financial need.

ORGANIZATIONAL EXPERIENCE

Himatif (Himpunan Mahasiswa Teknologi Informasi) - Vice Chairman | Aug 2023 – Jul 2024

AWARDS & CERTIFICATIONS

- Google Advanced Data Analytics - Coursera (Mar 2025)
- elevAlte with Dicoding Program - Microsoft (Mar 2025)
- Junior Web Developer, Vocational School Graduate Academy Program - Digital Talent Scholarship (Aug 2023)